

$$\begin{aligned} \max z &= 2x_1 - x_2 + 2x_3 \\ \text{s.t. } 2x_1 + x_2 &\leq 10 \\ x_1 + 2x_2 - 2x_3 &\leq 20 \\ x_2 + 2x_3 &\leq 5 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

(35p) Solve the given linear programming problem with tabular form of simplex method and write the optimal solution(s).

① LP in equation form.

$$\max z = 2x_1 - x_2 + 2x_3$$

$$2x_1 + x_2 + S_1 = 10$$

$$x_1 + 2x_2 - 2x_3 + S_2 = 20$$

$$x_2 + 2x_3 + S_3 = 5$$

$$x_1, x_2, x_3, S_1, S_2, S_3 \geq 0.$$

E.V. Starting b.d.s. $(S_1, S_2, S_3) = (10, 20, 5)$
 (-) in z-row.

Basic	x_1	x_2	x_3	S_1	S_2	S_3	RHS
max z	-2	1	-2	0	0	0	0
LV $\Rightarrow S_1$	2	1	0	1	0	0	10
S_2	1	2	-2	0	1	0	20
S_3	0	1	2	0	0	1	5

Ratio min $\Rightarrow 10/2=5 \Rightarrow$
 $20/1=20$
 zero value exist

Both x_1 and x_3 can be selected as entering variable.

I selected x_1 .

Basic	x_1	x_2	x_3	S_1	S_2	S_3	RHS
z	0	2	-2	1	0	0	10
x_1	1	1/2	0	1/2	0	0	5
S_2	0	3/2	-2	-1/2	1	0	15
S_3	0	1	2	0	0	1	5

Ratio $\Rightarrow$ LV.

$$(-) (x_1 \text{ New row}) : \begin{array}{cccccc} -1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & -5 \end{array}$$

$$S_2 \text{ row} \quad + \begin{array}{cccccc} 1 & 2 & -2 & 0 & 1 & 0 & 20 \\ \hline 0 & 3/2 & & & & & \end{array}$$

$$2 \cdot (X_1 - \text{row}) + 0 \cdot WZ = \text{New } Z.$$

$$\begin{array}{ccccccc} 2 & 1 & 0 & 1 & 0 & 0 & 10 \\ -2 & 1 & -2 & 0 & 0 & 0 & 0 \\ \hline 0 & 2 & -2 & 1 & 0 & 0 & 10 \end{array}$$

I am just adding S_2 and S_3 rows up. Is it it?

Basic	X_1	X_2	X_3	S_1	S_2	S_3	RHS
Z	0	2	-2	1	0	0	10
X_1	1	1/2	0	1/2	0	0	5
S_2	0	3/2	-2	-1/2	1	0	15
S_3	0	1	2	0	0	1	5
Z	0	3	0	1	0	0	15
X_1	1	1/2	0	1/2	0	0	5
S_2	0	5/2	0	-1/2	1	1	20
X_3	0	1/2	1	0	0	1/2	5/2

Basic variables.

Which variable is the E.V.?

No E.V. exists.

This shows us there is alt. opt.

Since all Z -row values are nonnegative, we reached the opt. sol.

Z	0	3	0	1	0	0	15
X_1	1	1/2	0	1/2	0	0	5
S_2	0	5/2	0	-1/2	1	1	20
X_3	0	1/2	1	0	0	1/2	5/2
Z	0	3	0	1	0	0	15
X_1	1	1/2	0	1/2	0	0	5
S_2	0	3/2	-2	-1/2	1	0	15

x_1	1	$1/2$	0	$1/2$	0	0	5
s_2	0	$3/2$	-2	$-1/2$	1	0	15
s_3	0	1	2	0	0	1	5

→ This solution is optimal also.

So, the first opt solution is:

$$A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 5/2 \end{bmatrix}.$$

the second opt. sol. is:

$$B \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}.$$

We have infinitely many solutions, we may write them by convex combinations $\alpha \cdot A + (1-\alpha) \cdot B$, where $\alpha \in [0,1]$.

$$\alpha \begin{bmatrix} 5 \\ 0 \\ 5/2 \end{bmatrix} + (1-\alpha) \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5/2 + 5 - 5/2 \\ 0 \\ 5/2 \alpha + 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 5/2 \alpha \end{bmatrix}$$

Final solution

$$\begin{aligned} \min z &= x_1 + 2x_2 \\ \text{s.t. } -x_1 + x_2 &\leq 2 \\ x_1 + 4x_2 &\geq 4 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

Q2) Solve the given linear programming problem with

- a) (15p.) the graphical method and write the optimal solution.
b) (20p.) the algebraic method and write the optimal solution.

1. LP in equation form.

$$\min z = x_1 + 2x_2 + M \cdot R_1$$

$$\begin{aligned} \text{s.t. } -x_1 + x_2 + S_1 &= 2 \\ x_1 + 4x_2 - S_2 + R_1 &= 4 \\ x_1, x_2, S_1, S_2, R_1 &\geq 0. \end{aligned}$$

! Preprocess ! $\rightarrow$ for artificial variables.
(S_1, R_1) is starting b.f.s

$$\rightarrow R_1 = 4 - x_1 - 4x_2 + S_2$$

substitute this equation in z .

$$z = x_1 + 2x_2 + M(4 - x_1 - 4x_2 - S_2)$$

$$z = (1 - 4M)x_1 + (2 - 4M)x_2 + 4M - MS_2$$

$$z + (4M - 1)x_1 + (4M - 2)x_2 + MS_2 = 4M$$

(+) $\Rightarrow$ E.V. $\uparrow$ Basic

	x_1	x_2	S_1	S_2	R_1	Sol
min z	$4M - 1$	$4M - 2$	0	M	0	$4M$
S_1	-1	1	1	0	0	2
R_1	1	4	0	-1	1	4
z	0	$2 - 12M$	0	$5M - 1$	$1 - 4M$	$4 - 12M$
S_1	0	5	1	-1	1	6
x_1	1	4	0	-1	1	4

the most (+). $\Rightarrow$ L.V.

$$\left(\begin{matrix} \text{New} \\ x_{1-\text{row}} \end{matrix} \right) (1 - 4M) + \text{old } z\text{-row} = \text{New } z\text{-row}.$$

$1-4M$	$4-16M$	0	$4M-1$	$1-4M$	$4-16M$
$4M-1$	$4M-2$	0	M	0	$4M$
0	$2-12M$	0	$5M-1$	$1-4M$	$4-12M$

This LP is Unbounded.

means that

$$z \rightarrow -\infty.$$

$$z = 4 - 12M.$$

$$M \rightarrow \infty \text{ then } z \rightarrow -\infty$$